Retrieval-Augmented Distributed Image Compression with Multi-Reference Side Information

Guojun Xu, Jianwen Xiang, *Member, IEEE*, Mingyang Zhang, Yaning Xie, Junwei Zhou, *Member, IEEE*

***Abstract*—Extremely low-bitrate distributed image compression (DIC) is crucial for bandwidth-constrained visual sensing and distributed cloud storage. At such low bitrates (<0.1 bpp), the transmitted representation retains little target information, making reconstruction particularly sensitive to the relevance and geometric consistency of the available side information (SI). Existing learning-based DIC methods are commonly developed with a pre-associated SI under matched training and evaluation distributions, whereas practical decoder-side visual collections may contain references with varying correlations to the target. To address these limitations, we propose RADIC, a retrieval-augmented DIC framework that replaces fixed paired SI with multiple references retrieved from a decoder-side visual gallery. Specifically, we construct a heterogeneous SI gallery and retrieve correlated references using the transmitted latent representation. To handle the geometric mismatch and varying relevance of retrieved references, we design a depth-aware geometric alignment module to align retrieved features through depth-guided coarse flow and residual refinement. A multi-reference cross-attention mechanism then selectively aggregates complementary information while suppressing irrelevant content. RADIC is trained using target images from KITTI General and evaluated on KITTI General, KITTI Stereo, Cityscapes, and InStereo2K without further fine-tuning. Experimental results demonstrate superior perceptual compression performance and improved robustness to variations in SI correlation. Our code is released at https://github.com/Mommqq/RADIC.**

***Index Terms*—Distributed image compression, side information, image retrieval, perceptual compression, extremely low bitrates.**

I. Introduction

LARGE-SCALE multi-view systems, such as autonomous driving [1], robotic perception [2], and immersive communication [3], continuously generate correlated visual observations from different viewpoints. Efficient transmission of these observations is challenging when each view is captured and encoded independently under limited bandwidth. Meanwhile, practical decoders, such as cloud servers and edge storage nodes, may maintain large collections of previously stored visual data that can provide additional information for reconstruction [4]. This setting motivates distributed image compression (DIC), where the target image is encoded independently and side information (SI) available at the decoder is used to assist reconstruction [5].

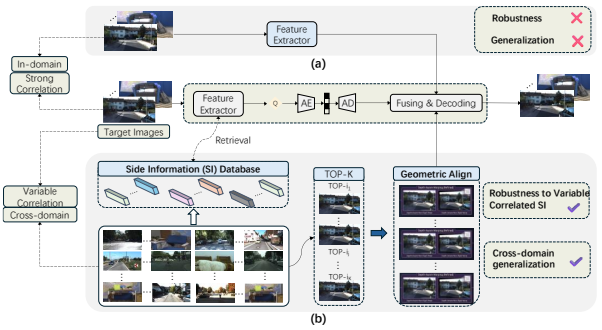


Fig. 1: Comparison of two DIC settings. (a) Existing methods [6], [7] rely on a paired SI for each target. (b) The proposed RADIC retrieves top-K references from a decoder-only visual database, aligns them geometrically, and uses the aligned features for reconstruction, thereby relaxing the paired-SI assumption.

As illustrated in Fig. 1, existing learning-based DIC methods are generally developed under a paired-SI setting, where each target is associated with a pre-matched stereo or multi-view reference [6]–[10]. In practical systems, however, the available SI may exhibit different levels of correlation with the target. As shown in Fig. 2, the same target can produce substantially different reconstructions when paired, non-paired, and cross-domain SI are provided to the decoder. As the correlation between target and SI decreases, larger differences in geometry and image content are introduced, making the SI less effective for reconstruction. In addition, target images may be acquired from different sensors and environments, resulting in changes in the source distribution. These effects become more significant at extremely low bitrates, where only limited target information is retained in the transmitted representation.

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Guojun Xu, Jianwen Xiang, Mingyang Zhang, Yaning Xie, and Junwei Zhou were with the School of Artificial Intelligence, Wuhan University of Technology, Wuhan, China, 430070. Corresponding author: Junwei Zhou (E-mail: junweizhou@msn.com).

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Fig. 2: Qualitative comparison of ATN, NDSC, MDIC, and RADIC under different decoder-only SI conditions. The top row shows the target image and the cross-domain (△), non-paired (◇), and paired (○) SI. For each method, the three reconstruction rows correspond to these SI conditions from top to bottom. The values below each reconstruction report Bpp/ PSNR/ LPIPS.

At extremely low bitrates, perceptual reconstruction becomes important because pixel-wise optimization tends to produce over-smoothed results when image details are largely removed from the transmitted representation [11]. Recent single-image compression methods [12], [13] have introduced generative priors to recover perceptually plausible structures and textures. MDIC [7] further extends perceptual reconstruction to DIC by exploiting multimodal SI. However, its visual SI is still provided as a pre-matched reference, and the reconstruction remains sensitive to the correlation between the target and the given SI.

To address these challenges, as illustrated in Fig. 1, we propose RADIC, a retrieval-augmented DIC framework for perceptual reconstruction at extremely low bitrates. At the decoder, the decoded target representation is used to retrieve the top-K references from a visual gallery containing SI from multiple domains. RADIC supports reconstruction with a single reference while allowing multiple references to be exploited when available, extending DIC beyond predefined reference pairs.

Retrieval from a heterogeneous gallery introduces substantial geometric mismatch and relevance variation among the candidate references. To establish the correspondence between target and references, we introduce a Depth-Aware Geometric Alignment (DAGA) module that combines depth-guided coarse flow with residual refinement. A Multi-reference Cross-Attention (MCA) mechanism then performs spatially selective aggregation, retaining complementary information while suppressing irrelevant content. To improve robustness to varying SI quality, decoder attention further supervises an encoder-side importance map during training, transferring SI-aware guidance to rate allocation. As shown in Fig. 2, RADIC preserves clearer scene structures under paired, non-paired, and cross-domain SI, showing improved robustness to changes in SI correlation.

Our main contributions are summarized as follows:

- We formulate a retrieval-augmented DIC setting in which multiple SI references are selected from a decoder-only visual gallery according to the received target representation, enabling perceptual reconstruction at extremely low bitrates under varying correlations between target and references.
- We address the geometric mismatch and varying relevance of retrieved SI through depth-aware alignment and selective multi-reference fusion, which exploit complementary information while suppressing irrelevant content. Decoder attention further guides importance learning for SI-aware rate allocation.

- RADIC is trained only on KITTI General and evaluated on KITTI General, KITTI Stereo, Cityscapes, and InStereo2K. Experiments show favorable perceptual performance across four test sets under varying SI conditions.

The remainder of this paper is organized as follows. Section II reviews related work on correlated-source image compression and extremely low-bitrate perceptual compression. Section III analyzes the theoretical motivation. Section IV presents the proposed RADIC framework. Section V describes the experimental settings and evaluates the proposed method in terms of compression performance and robustness. Section VI further analyzes the use of multiple SI references under different SI conditions. Finally, Section VII concludes the paper.

II. Related Work

A. Correlated-Source Image Compression

Correlated visual sources can be compressed through two main paradigms. Joint image compression (JIC) accesses correlated images at both the encoder and decoder, allowing direct inter-source interaction and joint entropy modeling. ECSIC [14] introduces epipolar-constrained stereo cross-attention and stereo context modules for entropy estimation. BiSIC [15] performs bidirectional stereo compression with a cross-dimensional entropy model to capture dependencies between stereo latents. CAMSIC [16] employs content-aware masked image modeling to estimate spatial and disparity dependencies. 3D-LMVIC [17] further incorporates 3D geometric priors for wide-baseline multi-view compression. Although these methods effectively exploit inter-source correlations, they rely on encoder-side interaction and are unsuitable for distributed scenarios.

Distributed image compression (DIC) encodes correlated sources independently and exploits their dependencies using side information (SI). Slepian–Wolf theory [18] shows that lossless distributed coding can achieve the joint-coding rate region, while Wyner–Ziv coding [19] extends this setting to lossy compression. In learning-based DIC, NDIC [5] extracts common information from SI and combines it with the transmitted target representation. ATN [10] uses cross-attention to align target and SI features. NDSC [6] formulates lossy distributed coding with a conditional VQ-VAE, while LDMIC [9] employs symmetric DIC for multi-view reconstruction. More recently, DIC-DDA [20] improves asymmetric DIC through dual-domain alignment between the decoded target and SI.

These methods achieve competitive rate-distortion performance on stereo and multi-view datasets, but most are optimized for pixel fidelity and produce blurred reconstructions at extremely low bitrates. They also commonly rely on a single paired SI image and in-domain training and evaluation. Consequently, their performance is sensitive to changes in SI correlation and limited in generalizing across datasets.

B. Extremely Low-Bitrate Perceptual Compression

At extremely low bitrates, perceptual optimization helps preserve richer semantic information and alleviate the blurring caused by severe information loss [13], [21], [22]. Early perceptual compression methods employ generative models to improve reconstruction quality at low rates [23]. More recent approaches make use of stronger semantic and diffusion priors. PerCo [11] conditions an iterative diffusion decoder on vector-quantized latents and a global description, while ResULIC [24] combines residual-guided semantic coding with compression-aware diffusion. MISC [25] introduces large multimodal models for ultra-low bitrate semantic compression. These methods focus on single-image compression and do not exploit decoder-only correlated sources.

Perceptual reconstruction has also been introduced into DIC. MDIC [7] exploits multimodal decoder-side SI for extremely low-bitrate reconstruction, while CODIC [26] considers distortion-perception optimization in distributed compression. However, both methods are developed with pre-associated SI and do not explicitly consider large variations in target–SI correlation. Their reconstruction remains dependent on the quality of the given SI. We address these limitations by retrieving multiple references from a heterogeneous multi-domain SI gallery and selectively exploiting correlated information while suppressing irrelevant content.

III. Information-Theoretic Analysis

A. Problem Formulation

Let $X^n = (X_1, \ldots, X_n)$ denote a target source block of length n, taking values in $\mathcal{X}^n$. Let $Y_{1:K}^n = (Y_1^n, \ldots, Y_K^n)$ denote K SI sources available only at the decoder. The encoder maps the target block to a message

$$f_n : \mathcal{X}^n \to \{1, \ldots, M_n\}, \tag{1}$$

where M_n is the number of possible messages and satisfies

$$\frac{1}{n} \log_2 M_n \le R. \tag{2}$$

For an observed source block X^n, the transmitted message is

$$M = f_n(X^n). \tag{3}$$

Here, R is the average number of transmitted bits per source symbol. Equivalently, one may set $M_n = \lfloor 2^{nR} \rfloor$.

The decoder reconstructs the target using the message and SI:

$$\widehat{X}^n = g_n(M, Y_1^n, \ldots, Y_K^n). \tag{4}$$

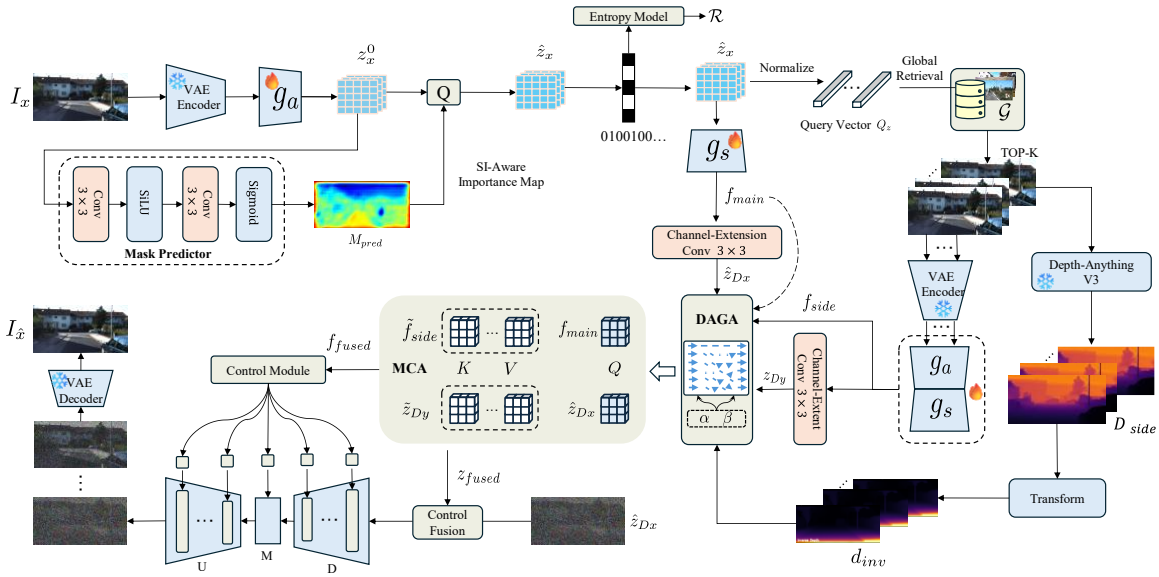


Fig. 3: Overall architecture of RADIC. The encoder maps the target image I_x to a latent representation z_x and predicts an SI-aware importance map M_{pred} for latent modulation before quantization. At the decoder, $\hat{z}_x$ retrieves top-K references from the gallery $\mathcal{G}$, which are aligned by Depth-Aware Geometric Alignment (DAGA) and screened by Multi-reference Cross-Attention (MCA). The fused reference features condition a latent diffusion model to reconstruct $\hat{I}_x$ at extremely low bitrates.

For a single-letter distortion measure $d(x, \hat{x})$, an (n, R, D) code satisfies

$$\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\Big[d(X_i, \widehat{X}_i)\Big] \le D. \tag{5}$$

B. Condition for Multi-Reference Gain

For lossy reconstruction, the Wyner–Ziv rate–distortion function with K decoder-only references is

$$R_{\text{WZ}}^{(K)}(D) = \min_{\substack{p(u|x),\, \hat{x}(u,y_{1:K}) \\ \mathbb{E}[d(X,\widehat{X})]\le D}} I(X; U \mid Y_{1:K}), \tag{6}$$

where U denotes the compressed representation generated from X, and

$$\widehat{X} = \hat{x}(U, Y_{1:K}). \tag{7}$$

The Markov chain

$$U - X - Y_{1:K} \tag{8}$$

means that U and $Y_{1:K}$ are conditionally independent given X:

$$p(u \mid x, y_{1:K}) = p(u \mid x). \tag{9}$$

This enforces the decoder-only SI constraint because the encoder generates U without observing any reference.

Define the optimal rate gain from adding the $(K+1)$-th reference as

$$\Delta_{K+1}(D) = R_{\text{WZ}}^{(K)}(D) - R_{\text{WZ}}^{(K+1)}(D). \tag{10}$$

The additional reference strictly improves compression efficiency if and only if

$$\Delta_{K+1}(D) > 0. \tag{11}$$

Equivalently, assuming that the minimum is attained, there must exist a feasible representation and reconstruction function for the $(K+1)$-reference problem such that

$$I(X; U \mid Y_{1:K}, Y_{K+1}) < R_{\text{WZ}}^{(K)}(D). \tag{12}$$

For a fixed representation U satisfying

$$U - X - (Y_{1:K}, Y_{K+1}), \tag{13}$$

the reduction in conditional mutual information is

$$\begin{aligned} &I(X; U \mid Y_{1:K}) - I(X; U \mid Y_{1:K}, Y_{K+1}) \\ &\quad = I(U; Y_{K+1} \mid Y_{1:K}). \end{aligned} \tag{14}$$

For the current compressed representation, the $(K+1)$-th reference yields a strict rate gain if and only if

$$I(U; Y_{K+1} \mid Y_{1:K}) > 0. \tag{15}$$

Eq. (15) means that Y_{K+1} must provide information about U that is not already contained in the first K references. The new reference is conditionally redundant and cannot further reduce the required rate for that representation when

$$I(U; Y_{K+1} \mid Y_{1:K}) = 0, \tag{16}$$

Hence, increasing K is beneficial only while newly added references provide complementary target information. As their conditional information decreases, the rate gain approaches zero and eventually saturates.

IV. Methodology

A. Overall Framework

The proposed RADIC framework consists of an independent encoder and a retrieval-augmented decoder, as illustrated in Fig. 3. Given a target image $I_x \in \mathbb{R}^{H\times W\times 3}$, a VAE encoder $\mathcal{E}_{vae}$ first maps it into a latent representation, which is further processed by an analysis transform g_a to obtain the compression latent z_x^0. Before quantization, an SI-aware Importance Map (IM) modulates z_x^0 according to the predicted availability of decoder-only SI. The resulting latent is quantized and entropy-coded into the transmitted bitstream.

At the decoder, the decoded latent $\hat{z}_x$ is used to retrieve the top-K candidate references from a decoder-only visual gallery $\mathcal{G}$. The retrieved references are aligned with the target representation by the Depth-Aware Geometric Alignment (DAGA) module. A Multi-reference Cross-Attention (MCA) module then performs spatial reference screening and feature fusion. The fused reference features, together with the decoded target representation, condition a latent diffusion model for reconstruction at extremely low bitrates.

B. Encoding and SI-Aware Rate Allocation

In the distributed coding setting, the encoder has no access to SI. It cannot determine which target regions can be recovered from the available SI. RADIC addresses this issue by learning an encoder-side importance map from the target representation and supervising it with decoder-side attention scores during training.

The target image is first mapped to the compression latent

$$z_x^0 = g_a(\mathcal{E}_{vae}(I_x))\,. \tag{17}$$

A mask predictor $\mathcal{P}_m$ then estimates a spatial IM

$$M_{pred} = \mathcal{P}_m(z_x^0), \tag{18}$$

where $M_{pred} \in [0,1]^{H_z\times W_z}$, and the modulated latent is

$$z_x = z_x^0 \odot (1 - \eta M_{pred})\,, \tag{19}$$

where $\odot$ denotes element-wise multiplication and η controls the attenuation strength.

C. Quantization and Entropy Modeling

Following RDEIC [13], a hybrid quantization scheme is adopted. The modulated latent z_x is quantized by rounding:

$$\hat{z}_x = Q(z_x). \tag{20}$$

A hyper-analysis transform h_a produces the hyper-latent

$$z_{h,x} = h_a(z_x), \tag{21}$$

which is quantized using a VQ codebook [27] to obtain $\hat{z}_{h,x}$.

For entropy coding, $\hat{z}_x$ is divided into ten channel slices, and each slice is partitioned into checkerboard anchor and non-anchor symbols. For a symbol $\hat{z}_{x,s,u}$ at position u in slice s, its Gaussian likelihood is conditioned on the quantized hyper-latent, the previously decoded slices, and the available checkerboard context. The corresponding rate is estimated as

$$R_{z_x} = -\frac{1}{HW}\sum_s\sum_u \log_2 p_{\psi_{s,u}}\left(\hat{z}_{x,s,u} \mid \hat{z}_{h,x}, \hat{z}_{x,<s}, \mathcal{C}_{s,u}\right), \tag{22}$$

where $\mathcal{C}_{s,u}$ denotes the available spatial context. Anchor symbols are decoded first and are subsequently used as context for non-anchor symbols.

The hyper-latent is quantized using a codebook $\mathcal{E}$ with $|\mathcal{E}| = 2^{14}$. Since each VQ index corresponds to a 64×64 region of the input image, its fixed rate contribution is

$$R_{z_h} = \frac{H_h W_h \log_2 |\mathcal{E}|}{HW} = \frac{14}{64^2} \simeq 0.003418 \text{ bpp}. \tag{23}$$

The bitrate is

$$\mathcal{R} = R_{z_x} + R_{z_h}. \tag{24}$$

The VQ branch is optimized with

$$\mathcal{L}_{\text{VQ}} = \|\text{sg}[z_{h,x}] - e\|_2^2 + \lambda_{\text{com}}\,\|z_{h,x} - \text{sg}[e]\|_2^2, \tag{25}$$

where e denotes the selected codebook embedding, λ_{com} controls the commitment term, and $\text{sg}[\cdot]$ denotes the stop-gradient operator.

D. Retrieval-Augmented Generative Decoding

Upon receiving the quantized latent $\hat{z}_x$, the decoder first reconstructs the target feature $f_{main} = g_s(\hat{z}_x)$, which is further compressed along the channel dimension to obtain the latent representation $\hat{z}_{Dx} = \mathcal{C}_c(f_{main})$. Consistent with the target encoding process, for each retrieved side image I_{y_k} ($k \in \{1,\dots,K\}$), we similarly extract $f_{side}^{(k)}$ and z_{Dy_k}.

1) Depth-Aware Geometric Alignment (DAGA). Multiple SI sources inevitably suffer from complex geometric distortions. To align these features, we propose the DAGA module (Fig. 4) driven by monocular depth priors.

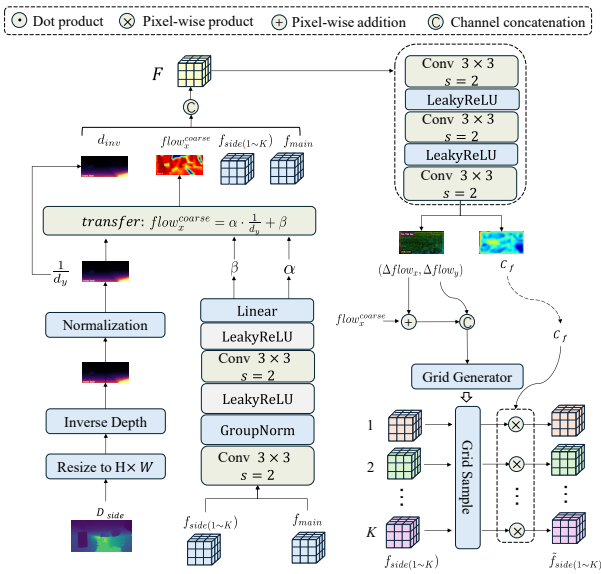


Fig. 4: Depth-Aware Geometric Alignment (DAGA). It predicts global affine parameters (α, β) to construct a depth-guided coarse flow, refines it with a residual flow, and warps the K side features. A learned confidence mask c_f is used to suppress unreliable regions.

For each side image, we extract its depth map D_{side} using the pre-trained Depth-Anything V3 [28]. The depth map is resized, inverted, and normalized to obtain the inverse depth d_{inv}. To model global viewpoint differences, a lightweight network predicts the affine parameters α and β from f_{main} and $f_{side}^{(k)}$. A coarse horizontal flow field is then generated by

$$flow_x^{coarse} = \alpha \cdot d_{inv} + \beta. \tag{26}$$

To refine the coarse alignment, a residual flow $\Delta flow = (\Delta flow_x, \Delta flow_y)$ and a confidence mask $c_f \in [0,1]^{H\times W}$ are estimated from d_{inv}, $flow_x^{coarse}$, $f_{side}^{(k)}$, and f_{main}. The final flow field is defined as

$$D_X = flow_x^{coarse} + \Delta flow_x, \;\; D_Y = \Delta flow_y. \tag{27}$$

The aligned side feature is then obtained by

$$\tilde{f}_{side}^{(k)} = \text{GridSample}\Big(f_{side}^{(k)}, (D_X, D_Y)\Big) \odot c_f, \tag{28}$$

where $\odot$ denotes pixel-wise multiplication.

2) Multi-reference Cross-Attention (MCA). Given the target feature f_{main} and the K aligned reference features $\tilde{f}_{side} \in \mathbb{R}^{B\times K\times C\times H\times W}$, MCA selects one reference at each spatial location according to the similarity between target and reference. The target feature is projected into a query, while the aligned reference features are projected into keys and values:

$$Q = W_Q f_{main}, \;\; K = W_K \tilde{f}_{side}, \;\; V = W_V \tilde{f}_{side}. \tag{29}$$

For each spatial location (i, j), the similarity score of the k-th reference is

$$s_k(i,j) = \frac{Q(i,j)\cdot K_k(i,j)}{\sqrt{C}}. \tag{30}$$

The hard selection mask is defined as

$$M_{Hard}^{(k)}(i,j) = \begin{cases} 1, & k = \arg\max_{k'} s_{k'}(i,j), \\ 0, & \text{otherwise.} \end{cases} \tag{31}$$

A soft mask is computed only for gradient propagation:

$$M_{Soft}^{(k)}(i,j) = \frac{\exp(s_k(i,j))}{\sum_{k'} \exp(s_{k'}(i,j))}. \tag{32}$$

The straight-through mask is

$$M_{STE}^{(k)} = M_{Soft}^{(k)} + \text{sg}\Big(M_{Hard}^{(k)} - M_{Soft}^{(k)}\Big), \tag{33}$$

Thus, the forward pass uses the one-hot hard mask, while gradients are propagated through $M_{Soft}^{(k)}$.

The selected reference feature is

$$f_{fused}(i,j) = \sum_{k=1}^{K} M_{Ste}^{(k)}(i,j)\cdot V_k(i,j). \tag{34}$$

The same MCA operation is applied to the aligned latent representations:

$$z_{fused} = \text{MCA}\left(z_{Dx}, \{\tilde{z}_{Dy_k}\}_{k=1}^{K}\right). \tag{35}$$

3) Feature and Latent Fusion. The fused representations are injected into the decoding process through residual modulation:

$$\tilde{f}_{main} = f_{main} + \phi_f([f_{main}, f_{fused}]), \tag{36}$$

$$\tilde{z}_{Dx} = \phi_g([\hat{z}_{Dx}, z_{fused}]) \odot \hat{z}_{Dx} + \phi_z([\hat{z}_{Dx}, z_{fused}]), \tag{37}$$

where ϕ_f, ϕ_z denote learnable fusion networks and $\phi_g(\cdot)$ is a gating function.

4) Decoding via Latent Diffusion Model. Inspired by RDEIC [13], we adopt relay residual diffusion for latent reconstruction. Let $x_0 = \mathcal{E}_{vae}(I_x)$ denote the target VAE latent, and let $\tilde{z}_{Dx}$ be the SI-enhanced compressed latent. Their residual is

$$e_x = \tilde{z}_{Dx} - x_0. \tag{38}$$

Instead of starting from pure noise, the diffusion trajectory is initialized from the compressed latent:

$$x_N = \sqrt{\bar{\alpha}_N}\tilde{z}_{Dx} + \sqrt{1-\bar{\alpha}_N}\epsilon_N, \qquad \epsilon_N \sim \mathcal{N}(0, I). \tag{39}$$

To make the trajectory agree with the compressed-latent initialization at $n = N$, we use $\lambda = (\bar{\alpha}_N^{-1} - 1)^{-1/2} = \sqrt{\bar{\alpha}_N/(1-\bar{\alpha}_N)}$. The relay residual diffusion process is defined as

$$x_n = \sqrt{\bar{\alpha}_n}x_0 + \sqrt{1-\bar{\alpha}_n}\left(\lambda e_x + \epsilon_n\right), \tag{40}$$

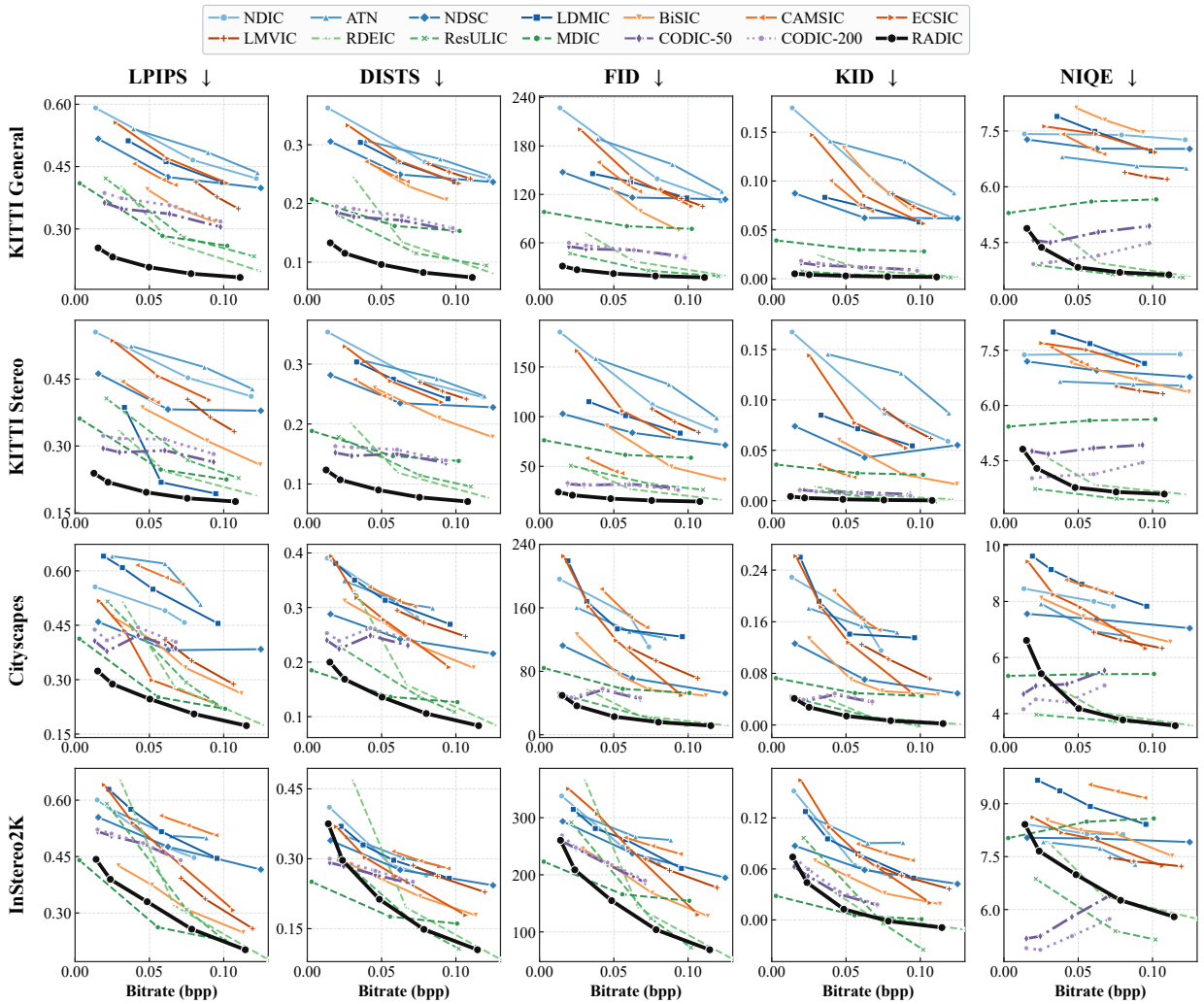


Fig. 5: Rate-perception comparisons on four test datasets. Lower (↓) is better for all metrics.

where the denoiser predicts the combined residual noise conditioned on the fused feature $\tilde{f}_{main}$:

$$\epsilon_\theta\left(x_n, n \mid \tilde{f}_{main}\right). \tag{41}$$

Starting from x_N, the reverse process estimates $\hat{x}_0$, which is decoded as

$$\hat{I}_x = \mathcal{D}_{\text{vae}}(\hat{x}_0). \tag{42}$$

E. Loss Functions

The overall training objective is defined as

$$\mathcal{L} = \lambda_1\mathcal{L}_{diff} + \lambda_2\mathcal{L}_{rate} + \lambda_3\mathcal{L}_{guide} + \lambda_4\mathcal{L}_{IM}. \tag{43}$$

Following the relay residual diffusion process, the diffusion loss supervises the prediction of the combined compression residual and Gaussian noise:

$$\mathcal{L}_{diff} = \omega_n\mathbb{E}\left[\left\|\lambda e_x + \epsilon_n - \epsilon_\theta\left(x_n, n \mid \tilde{f}_{main}\right)\right\|_2^2\right], \tag{44}$$

where $e_x = \tilde{z}_{Dx} - x_0$ and ω_n is the timestep-dependent weighting coefficient.

The rate loss combines the estimated coding rate and the vector-quantization objective:

$$\mathcal{L}_{rate} = \mathcal{R} + \mathcal{L}_{VQ}. \tag{45}$$

The latent guidance loss constrains the SI-enhanced compressed latent to remain close to the target VAE latent:

$$\mathcal{L}_{guide} = \|\tilde{z}_{Dx} - x_0\|_2^2. \tag{46}$$

To supervise the encoder-side importance map, we derive a spatial confidence map from the MCA attention weights and DAGA confidence masks:

$$A(i,j) = \begin{cases} 0, & K = 0, \\ \max\limits_{1\le k\le K} M_{Soft}^{(k)}(i,j)c_f^{(k)}(i,j), & K > 0. \end{cases} \tag{47}$$

The map characterizes the confidence of the selected SI at each spatial location and provides guidance for SI-aware rate allocation. The mask alignment loss is then defined as

$$\mathcal{L}_{IM} = \|M_{pred} - \mathrm{sg}(A)\|_2^2, \tag{48}$$

where $\mathrm{sg}(\cdot)$ denotes the stop-gradient operation.

V. Experimental Results

A. Experimental Settings

1) Datasets. RADIC is trained on KITTI General and evaluated on KITTI General [10], KITTI Stereo [29], Cityscapes [30], and InStereo2K [31]. KITTI General contains 174936 training pairs, 912 validation pairs, and 3607 test pairs. The test sets contain 790 KITTI Stereo pairs, 1525 Cityscapes pairs, and 50 InStereo2K pairs.
2) Decoder-Side SI Settings. During training, we construct a heterogeneous SI gallery by combining the training references from KITTI General, KITTI Stereo, Cityscapes, and InStereo2K. Target images are exclusively drawn from the KITTI General training set.

The decoded target latent is used to query the corresponding FAISS index. During training, the number of SI references is sampled independently for each input as $K \sim \mathrm{Uniform}\{0, 1, 2, 3\}$. For $K > 0$, FAISS retrieves the top-20 most similar gallery images, from which K references are randomly sampled.

During evaluation, a separate gallery is constructed using test references from the four datasets. For the main performance comparisons, RADIC is evaluated with a single paired SI reference ($K = 1$), consistent with the SI setting of existing DIC methods. We further evaluate multiple references to investigate the effect of additional SI on reconstruction performance. Under paired SI, the corresponding reference is used for all applicable DIC methods. For robustness evaluation, non-paired SI excludes the paired reference and all references from the same scene or sequence, while cross-domain SI excludes references from the target domain. KITTI General and KITTI Stereo are treated as the same domain.
3) Implementation Details. All images are center-cropped to 370×740 and resized to 256×512. Depth Anything V3 [28] is used for monocular depth estimation, while FAISS [32] is used for decoder-side retrieval. The diffusion decoder follows RDEIC with two denoising steps.

RADIC is trained using AdamW with an initial learning rate of 1×10^{-4}, and a weight decay of 1×10^{-2}. Cosine annealing is applied with a minimum learning rate of 1×10^{-6}. Each operating point is trained for 15 epochs with a batch size of 8 and a fixed random seed of 231. The loss coefficients are set to $\lambda_1 = 2.0$, $\lambda_3 = 2.0$, and $\lambda_4 = 0.1$, while $\lambda_2 \in \{16, 8, 4, 2, 1\}$ is varied to obtain different rate points. All experiments are conducted on NVIDIA L40 GPUs.

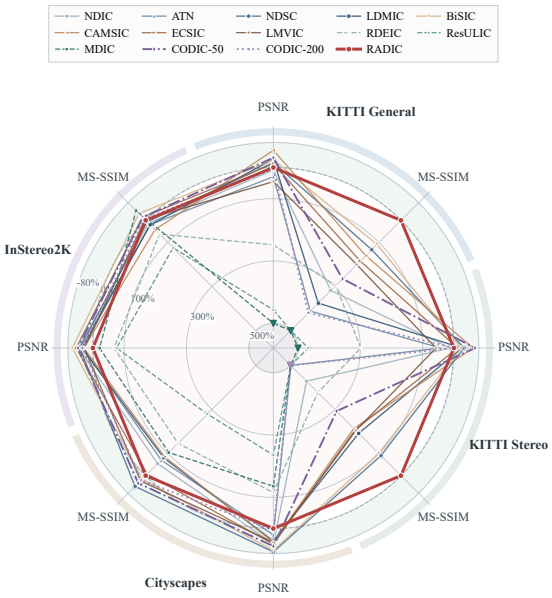


Fig. 6: Comparison of BD-Rate (%) [40], [41] computed with PSNR and MS-SSIM as quality metrics on four test datasets. RADIC is used as the anchor, and lower BD-Rate indicates better rate-distortion performance.

4) Evaluation Metrics. We use LPIPS [33], DISTS [34], FID [35], KID [36], and NIQE [37] for perceptual quality, and PSNR [38] and MS-SSIM [39] for pixel-level and structural fidelity. Perceptual metrics are used as the primary criteria at extremely low bitrates [11].
5) Baselines. We compare RADIC with the DIC methods NDIC [5], ATN [10], NDSC [6], LDMIC [9], MDIC [7], and CODIC [26]. We also include the joint multi-view codecs ECSIC [14], BiSIC [15], CAMSIC [16], and LMVIC [17], together with the single-image perceptual codecs RDEIC [13] and ResULIC [24]. All baselines are retrained on KITTI General.

B. Main Results

1) Quantitative Comparison. Fig. 5 compares perceptual performance at extremely low bitrates. RADIC achieves the best or near-best results on most perceptual metrics and maintains this advantage on KITTI Stereo, Cityscapes, and InStereo2K without dataset-specific fine-tuning. Under the common KITTI General training protocol, these results show favorable fixed-model transfer. Although CODIC achieves better NIQE scores at some low bitrates, its inferior full-reference perceptual metrics suggest a stronger reliance on generative priors, producing visually natural reconstructions with lower semantic consistency to the original images.

The perceptual gains do not consistently yield higher PSNR, reflecting the perception-distortion trade-off [42],

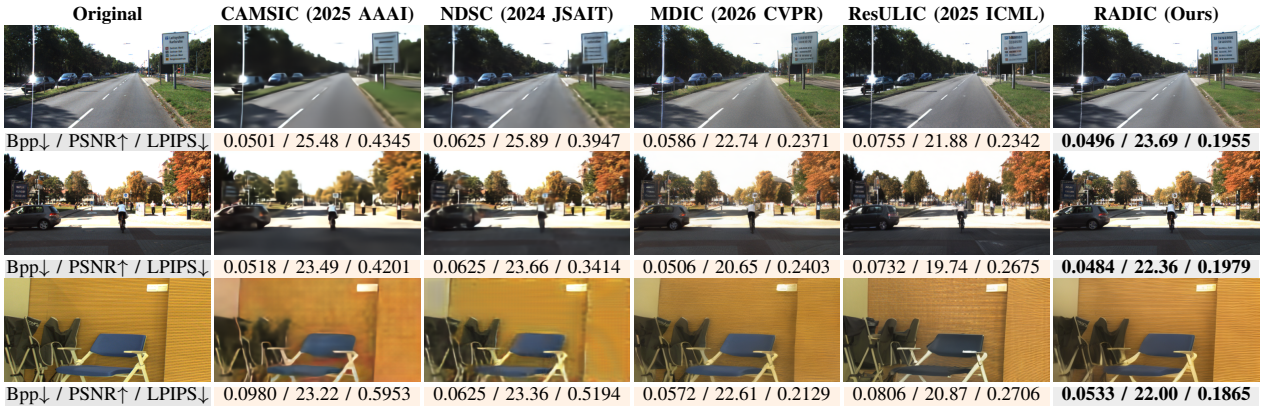


Fig. 7: Qualitative comparison among CAMSIC [16], NDSC [6], MDIC [7], ResULIC [24], and RADIC. CAMSIC and NDSC exhibit severe degradation at extremely low bitrates, while RADIC achieves the lowest LPIPS.

TABLE I

BD-PSNR AND BD-LPIPS UNDER PAIRED, NON-PAIRED, AND CROSS-DOMAIN RETRIEVAL. RADIC WITH PAIRED SI AT $K = 1$ SERVES AS THE COMMON ANCHOR ACROSS DATASETS. HIGHER BD-PSNR AND LOWER BD-LPIPS ARE BETTER, WHILE BOLD DENOTES THE BEST RESULT IN EACH COLUMN.

Domain	Method	K	Paired		Non-paired		Cross-domain	
			BD-PSNR ↑	BD-LPIPS ↓	BD-PSNR ↑	BD-LPIPS ↓	BD-PSNR ↑	BD-LPIPS ↓
KITTI General	NDSC	1	**1.26**	0.1533	-1.19	0.2985	-1.23	0.3133
	ATN	1	0.13	0.2824	**-0.29**	0.3096	**-0.34**	0.3135
	MDIC	1	-2.49	0.0665	-4.27	0.1675	-4.32	0.1807
	RADIC	1	0.00	0.0000	-1.03	0.0233	-1.12	0.0471
		2	0.06	-0.0031	-1.03	**0.0231**	-1.12	0.0428
		3	0.08	**-0.0043**	-1.03	0.0233	-1.13	**0.0401**
KITTI Stereo	NDSC	1	**0.27**	0.2121	-1.09	0.2899	-2.35	0.3608
	ATN	1	-0.48	0.3076	**-0.66**	0.3180	-1.17	0.3419
	MDIC	1	-3.61	0.1112	-4.70	0.1638	-5.48	0.2259
	RADIC	1	0.00	0.0000	-0.86	0.0515	**-0.95**	0.0563
		2	0.08	-0.0014	-0.87	**0.0512**	**-0.95**	**0.0561**
		3	0.17	**-0.0020**	-0.86	**0.0512**	**-0.95**	0.0562
Cityscapes	NDSC	1	**2.10**	0.1458	-0.98	0.2888	-2.18	0.3660
	ATN	1	-0.42	0.3475	**-0.64**	0.3522	**-0.92**	0.3880
	MDIC	1	-3.24	0.0584	-5.44	0.1541	-6.09	0.2023
	RADIC	1	0.00	**0.0000**	-2.39	0.1291	-2.45	0.1375
		2	-0.16	0.0104	-2.38	**0.1288**	-2.44	**0.1369**
		3	-0.23	0.0135	-2.39	0.1304	-2.43	0.1371
InStereo2K	NDSC	1	**1.05**	0.1637	-1.70	0.2619	-2.22	0.2680
	ATN	1	0.77	0.2169	**-0.21**	0.2544	**-0.35**	0.2596
	MDIC	1	-1.94	0.0183	-3.99	0.1087	-4.75	0.1194
	RADIC	1	0.00	0.0000	-1.61	0.0812	-1.66	**0.0816**
		2	-0.18	**-0.0074**	-1.61	**0.0808**	-1.66	0.0819
		3	-0.28	0.0137	-1.61	0.0814	-1.67	0.0833

[43]. As shown in Fig. 6, RADIC remains below several distortion-oriented methods in PSNR, particularly at higher rates, but achieves the best MS-SSIM on KITTI General and KITTI Stereo and remains competitive on Cityscapes and InStereo2K. This indicates that the perceptual improvements largely preserve structural fidelity despite lower pixel accuracy.

Compared with MDIC, RADIC achieves improved perceptual reconstruction under paired SI and further supports retrieval from a decoder-only visual gallery. Compared with RDEIC and ResULIC, SI provides structural and textural information unavailable from the compressed target alone.

2) Qualitative Comparison. As shown in Fig. 7, NDSC and CAMSIC exhibit severe structural distortion and texture loss at extremely low bitrates. MDIC and ResULIC recover more realistic content but introduce noticeable deviations from the original image. In contrast, RADIC preserves clearer structures and finer textures at a lower bitrate, achieving the lowest LPIPS.

3) Robustness to SI Correlation. Table I compares the robustness of different DIC methods under paired, non-paired, and cross-domain SI. For a fair comparison, each baseline uses the same top-1 retrieved SI as RADIC.

TABLE II
BD-RATE (%) ↓ OF RADIC (K=3) VARIANTS COMPUTED USING LPIPS, DISTS, PSNR, AND MS-SSIM. THE FULL RADIC MODEL IS USED AS THE ANCHOR.

Dataset	Variant	Components				BD-Rate (%) ↓			
		Ref.	DAGA	MCA	IM	LPIPS	DISTS	PSNR	MS-SSIM
KITTI General	C1	–	–	–	–	166.8	134.3	163.5	175.4
	C2	✓	–	–	–	81.0	65.8	87.2	88.5
	C3	✓	✓	–	–	13.8	11.5	14.9	14.9
	C4	✓	✓	✓	–	1.3	**-1.8**	11.3	1.8
	RADIC	✓	✓	✓	✓	**0.0**	0.0	**0.0**	**0.0**
KITTI Stereo	C1	–	–	–	–	182.1	127.5	189.7	197.9
	C2	✓	–	–	–	95.4	81.4	95.0	97.0
	C3	✓	✓	–	–	15.7	13.4	18.0	16.2
	C4	✓	✓	✓	–	11.3	12.5	11.2	13.9
	RADIC	✓	✓	✓	✓	**0.0**	**0.0**	**0.0**	**0.0**
Cityscapes	C1	–	–	–	–	38.7	38.6	67.7	115.6
	C2	✓	–	–	–	36.6	35.0	58.7	110.8
	C3	✓	✓	–	–	27.3	27.6	50.1	65.8
	C4	✓	✓	✓	–	1.1	0.7	31.5	0.5
	RADIC	✓	✓	✓	✓	**0.0**	**0.0**	**0.0**	**0.0**
InStereo2K	C1	–	–	–	–	132.8	131.7	94.8	172.2
	C2	✓	–	–	–	38.8	30.9	45.2	81.2
	C3	✓	✓	–	–	10.6	15.9	18.9	14.5
	C4	✓	✓	✓	–	6.6	4.0	10.7	10.9
	RADIC	✓	✓	✓	✓	**0.0**	**0.0**	**0.0**	**0.0**

RADIC achieves the lowest BD-LPIPS across all datasets and retrieval settings, showing that its perceptual reconstruction remains stable as SI correlation decreases. The transition from paired to cross-domain SI causes substantially smaller degradation than MDIC in both BD-LPIPS and BD-PSNR across the four datasets. This indicates that DAGA and MCA reduce the effect of geometric mismatch and irrelevant content. ATN is generally more robust than NDSC in BD-PSNR and in several cross-domain perceptual comparisons, which is consistent with its use of cross-attention for feature selection, although it remains notably below RADIC in BD-LPIPS. Increasing K from 1 to 3 produces only marginal changes under non-paired and cross-domain retrieval, indicating that the first retrieved reference provides most of the available information and that additional weakly related candidates are largely suppressed.

4) Complexity Analysis. Table III compares cached and online depth processing for paired SI ($K = 1$). Online decoding includes Depth Anything (0.0885 ± 0.0066 s); cached decoding reuses reference depth. Both recompute reference features. Entropy-model computation belongs to encoding; arithmetic coding is excluded.

TABLE III
RUNTIME (MEAN ± SD) AND BD-RATE ON CITYSCAPES. ANCHOR: RADIC ($K = 1$).

Model	Time (s)		BD-Rate (%) ↓	
	Enc.	Dec.	LPIPS	PSNR
ResULIC	0.0749 ± 0.0136	0.5143 ± 0.0158	144.42	262.77
RDEIC	0.0149 ± 0.0001	0.0689 ± 0.0001	105.85	174.94
MDIC	0.0331 ± 0.0026	0.4244 ± 0.0128	4.25	201.95
RADIC (cached)	0.0206 ± 0.0007	0.1245 ± 0.0003	0.00	0.00
RADIC (online)	0.0206 ± 0.0007	0.2157 ± 0.0139		

C. Ablation Studies

Table II reports the cumulative ablation results of RADIC. C_1 denotes the base codec. Reference retrieval, DAGA, and MCA are introduced in C_2, C_3, and C_4, while the full model further includes the importance map.

1) Reference Retrieval. Adding retrieved SI from C_1 to C_2 reduces BD-Rate for all four metrics on all datasets. The improvements are substantial on KITTI General, KITTI Stereo, and InStereo2K. The gain is smaller on Cityscapes, indicating that the benefit of retrieval varies with the available SI. These results show that retrieved references provide useful information beyond the compressed target.

2) Depth-Aware Geometric Alignment. Introducing DAGA in C_3 further reduces BD-Rate across all datasets and metrics. The reductions are particularly large on KITTI General, KITTI Stereo, and InStereo2K. Cityscapes also shows clear improvement, especially in MS-SSIM. The consistent gains show the importance of reducing geometric differences between the target and retrieved references before feature fusion.

3) Multi-reference Cross-Attention. Adding MCA in C_4 provides further gains after geometric alignment. The

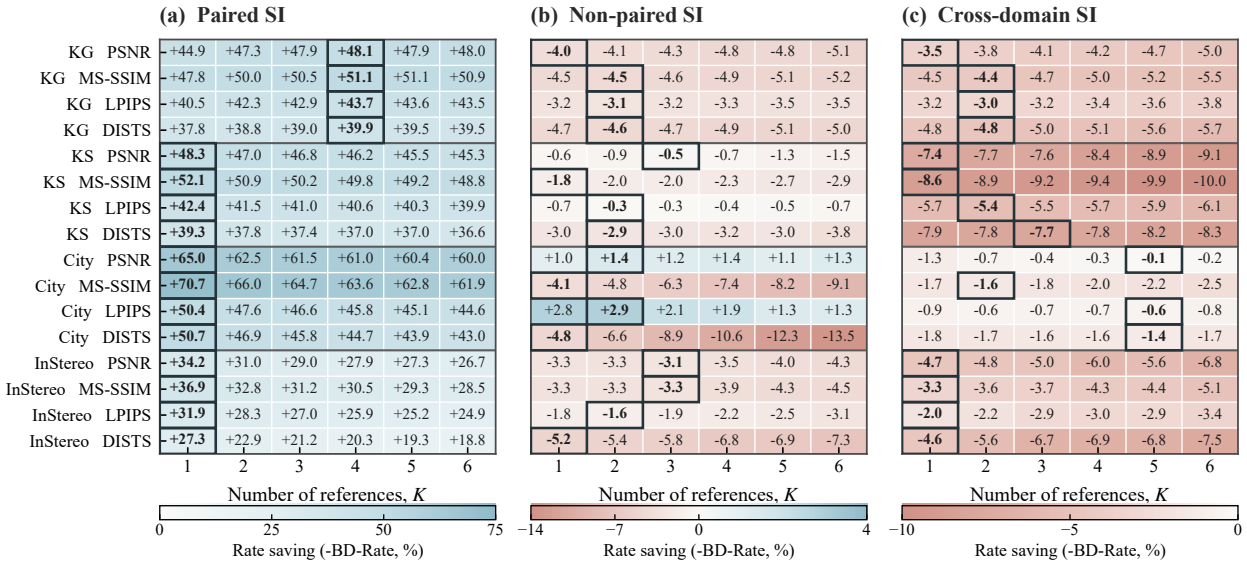


Fig. 8: Effect of the number of SI references under paired, non-paired, and cross-domain SI. Each cell reports the rate saving ($-$BD-Rate, %) relative to $K = 0$ under the same SI setting, while blue and red indicate gains and losses, respectively, and larger values indicate better coding efficiency. Bold outlined cells mark the best K. KG and KS denote KITTI General and KITTI Stereo.

largest improvements are observed on Cityscapes and KITTI General, while the changes on KITTI Stereo and InStereo2K are smaller. This result is consistent with the role of MCA in selecting and combining information from the aligned references.

4) SI-Aware Rate Allocation. The full RADIC model improves most metric combinations over C_4. Clear gains are observed across all four metrics on KITTI Stereo and InStereo2K, while the PSNR improvement is especially large on Cityscapes. On KITTI General, C_4 achieves a slightly lower BD-Rate for DISTS, with a difference of 1.8%. The overall results show that the importance map provides additional gains through adaptive modulation of the transmitted latent.

VI. Discussion

The ablation results show that retrieved SI improves compression performance, and that DAGA and MCA provide further gains after retrieval. We next examine how the number of references affects the results under different SI conditions.

Fig. 8 reports the rate saving from $K = 0$ to $K = 6$ under paired, non-paired, and cross-domain SI. Rate saving is defined as the negative BD-Rate relative to $K = 0$ under the same SI setting. PSNR provides the closest empirical comparison with the distortion term in the information-theoretic analysis because it is directly related to MSE for a fixed image range.

Under paired SI, the first reference provides most of the compression gain across all four datasets. KITTI General benefits further from additional references, as its sequential driving scenes provide multiple correlated observations with complementary visual information. This observation is consistent with the condition

$$I(U; Y_{K+1} \mid Y_{1:K}) > 0, \tag{49}$$

which describes whether an additional reference contains information not already provided by the existing SI.

In contrast, the gains on KITTI Stereo gradually decrease with increasing K. For Cityscapes and InStereo2K, the evaluation galleries are constructed from stereo image pairs, where the matched reference provides most of the useful information. Additional retrieved references therefore yield limited improvements.

Under non-paired and cross-domain SI, the benefits of additional references are further reduced due to weaker correlations with the target. We use $K = 1$ for the main comparisons with existing DIC methods and investigate larger values of K to evaluate the ability of RADIC to exploit complementary SI from multiple references.

Furthermore, PSNR, LPIPS, and DISTS do not always follow the same trend. LPIPS and DISTS evaluate perceptual similarity in feature space, while PSNR is derived from pixel-level distortion.

VII. Conclusion

We present RADIC, a distributed image compression framework that retrieves one or multiple side information (SI) references from a decoder-side visual gallery

for extremely low-bitrate reconstruction. Retrieved references are aligned by the Depth-Aware Geometric Alignment (DAGA) module and fused by the Multi-Reference Cross-Attention (MCA) module to handle geometric and content differences. Decoder attention is also used to supervise an encoder-side importance map for SI-aware rate allocation, while the encoder uses only the target image. Experiments on KITTI General, KITTI Stereo, Cityscapes, and InStereo2K show that RADIC achieves strong perceptual compression performance with a single reference and improved robustness under varying SI correlations. The analysis of the number of references shows that most of the gain is obtained from a small number of relevant references. Additional references provide limited benefit when their correlation with the target becomes weak. These results demonstrate the effectiveness of retrieved SI for distributed image compression and highlight the importance of reference relevance in decoder-side visual galleries.

Limitations. Although RADIC is more efficient than existing perceptual DIC methods, further reducing retrieval and diffusion costs remains important. As a new compression paradigm between cloud and edge, broader exploration of distributed data in open scenarios is needed to support scalable deployment.

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Guojun Xu is currently pursuing a Ph.D. degree with the School of Artificial Intelligence, Wuhan University of Technology. His current research interests include distributed source coding, image compression, and other visual problems.



Jianwen Xiang (Member, IEEE) received the Ph.D. degree in computer software and theory from Wuhan University and in information science from the Japan Advanced Institute of Science and Technology. Since then, he has been a Special Research Fellow with the National Institute of Advanced Industrial Science and Technology, Japan, the Director of the Central Research Laboratory, NEC Corporation, Japan, and a Postdoctoral Researcher with the Japan Advanced Institute of Science and Technology. He is currently a Professor with the School of Artificial Intelligence, Wuhan University of Technology, China, and a Visiting Professor with Kyushu University, Japan. His research interests include dependable computing, information security, and formal methods. He has served as a Reviewer for international journals, including IEEE Transactions on Reliability, Reliability Engineering and System Safety, ACM Transactions on Software Engineering and Methodology, and Journal of Systems and Software.



Mingyang Zhang is currently pursuing the M.S. degree with the School of Artificial Intelligence, Wuhan University of Technology. His current research interests include distributed source coding and multi-reference face super-resolution.



Yaning Xie is currently pursuing the M.S. degree with the School of Artificial Intelligence, Wuhan University of Technology. His current research interests include distributed source coding, lossless image compression, and computer vision.



Junwei Zhou (Member, IEEE) received the Ph.D. degree from the Department of Electrical Engineering, City University of Hong Kong. Since then, he has been a Visiting Scholar with the Polytechnic University of Turin, Italy, and a Researcher with the City University of Hong Kong, Hong Kong, China. He is currently a Professor with the School of Artificial Intelligence, Wuhan University of Technology, China, and a Postdoctoral Researcher with Pennsylvania State University, USA. His research interests include computer vision, information security, and distributed source coding. He has served as a PC Member for conferences including EMNLP 2023, Coling 2022, ACL 2022, and ISSRE 2022.